## MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. $(2;1)\cdot(3;-4)$ equals

   A. $2$
   B. $10$
   C. $(6;-4)$
   D. $-2$

2. The norm of ( $-3$; 4) is

   A. $7$
   B. $25$
   C. $5$
   D. $1$

3. Two vectors whose scalar product is zero are

   A. equal
   B. collinear
   C. opposite
   D. orthogonal

4. Norms 2 and 3, angle 60 $^\circ$ : product

   A. $6$
   B. $3$
   C. $\sqrt{3}$
   D. $0$

5. $\|u-v\|^2$ is

   A. $\|u\|^2-\|v\|^2$
   B. $\|u\|^2+\|v\|^2$
   C. $\|u\|^2-2u\cdot v+\|v\|^2$
   D. $\|u\|^2+2u\cdot v+\|v\|^2$

## MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. An obtuse angle between non-zero vectors gives a scalar product that is

   A. positive
   B. zero
   C. equal to 1
   D. negative

2. In an orthonormal basis, $u\cdot i$ is

   A. the x-coordinate of u
   B. the norm of u

C. the y-coordinate of u

D. always zero

3. The locus $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for $A \neq B$ is

A. the perp. bisector

B. the line AB

C. the circle with diameter AB

D. the midpoint alone

4. If $u \cdot v = 3$, $(2u) \cdot (-v)$ is

A. $-3$

B. $-6$

C. $6$

D. $9$

5. For $AB = AC = 2$ and a right angle at A, $BC$ is

A. $4$

B. $2$

C. $8$

D. $2\sqrt{2}$

## True or false? Justify.

1. The scalar product of two vectors is always positive.

2. The zero vector is orthogonal to every vector.

3. $u \cdot v = 0$ implies that one of the vectors is zero.

**Personal review**

I distinguish norm and squared norm; I choose a suitable formula; I check the coordinate system; I write an equivalence for a locus.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.